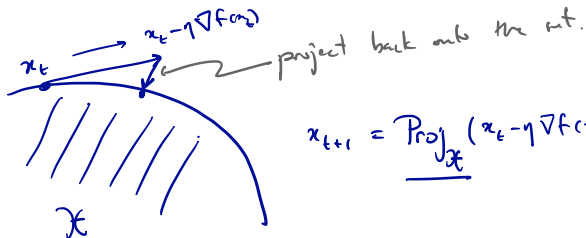


Projections and Proximal Operators

$$\min: f(x)$$

$$\text{st: } x \in \mathcal{X}$$



$$x_{t+1} = \text{Proj}_{\mathcal{X}}(x_t - \eta \nabla f(x_t))$$

Projected subgradient method

$$\begin{aligned} \min: & f(x) \\ \text{s.t.: } & x \in \mathcal{X} \end{aligned} \quad \begin{aligned} & f \text{ convex and } \|g_x\| \leq G, \quad \forall g_x \in \partial f(x) \\ & x_{t+1} = \text{Proj}_{\mathcal{X}}(y_{t+1}) \\ & y_{t+1} = x_t - \eta g_t, \quad g_t \in \partial f(x_t) \end{aligned}$$

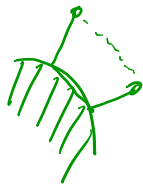
$$\begin{aligned} \|x_{t+1} - x^*\|^2 &= \|x_t - \eta g_t - x^*\|^2 \\ &= \|x_t - x^*\|^2 - 2\eta \langle g_t, x_t - x^* \rangle + \eta^2 \|g_t\|^2 \\ &\leq \|x_t - x^*\|^2 - 2\eta (f(x_t) - f(x^*)) + \eta^2 G^2 \end{aligned}$$

$$f(x_t) - f(x^*) \leq \frac{1}{2\eta} \left(\|x_t - x^*\|^2 - \|x_{t+1} - x^*\|^2 \right) + \frac{\eta}{2} G^2$$

$$\left(\begin{aligned} x_{t+1} &= \text{Proj}_{\mathcal{X}}(y_{t+1}) \\ x^* &= \text{Proj}_{\mathcal{X}}(x^*) \end{aligned} \quad \|y_{t+1} - x^*\|^2 \geq \|x_{t+1} - x^*\|^2 \right)$$

$$\leq \frac{1}{2\eta} \left(\|x_t - x^*\|^2 - \|x_{t+1} - x^*\|^2 \right) + \frac{\eta}{2} G^2$$

$$\Rightarrow f\left(\frac{1}{T} \sum x_t\right) - f(x^*) \leq \frac{1}{2\eta} T \cdot R^2 + \frac{\eta}{2} G^2 \longrightarrow \frac{1}{\sqrt{T}} \text{ convergent}$$



Projection and the Prox operator

$$\begin{aligned} \min: & f(x) \\ \text{st. } & x \in \mathcal{X} \end{aligned}$$

$$\longleftrightarrow \min: f(x) + I_{\mathcal{X}}(x),$$

could minimize using
subgradient method

$I_{\mathcal{X}}(x)$ - "indicator fn"

$$I_{\mathcal{X}}(x) = \begin{cases} +\infty & x \notin \mathcal{X} \\ 0 & x \in \mathcal{X} \end{cases}$$

Note: Proj. sub-gr vs sub-gr on $(f(x) + I_{\mathcal{X}}(x))$ convex fn.

$$\begin{aligned} \text{Proj}_{\mathcal{X}}(x) &= \arg \min_u: I_{\mathcal{X}}(u) + \frac{1}{2} \|u - x\|_2^2 \\ &= \arg \min_u: \frac{1}{2} \|u - x\|_2^2 \\ &\quad \text{st. } u \in \mathcal{X} \end{aligned} \quad \left. \begin{array}{l} \text{generalize to other} \\ \text{convex functions} \end{array} \right\} = \text{Proj}_{\mathcal{X}}(x).$$

Def'n: For h a convex function

$$\text{prox}_h(y) = \arg \min_x: h(x) + \frac{1}{2} \|x - y\|_2^2$$

$$\text{prox}_{\eta h}(y) = \arg \min_x: h(x) + \frac{1}{2\eta} \|x - y\|_2^2$$

Proximal Gradient Algorithm

moving away from the oracle model of computation

$$\begin{aligned} \text{Proj. GD : } x_{t+1} &= \text{Proj}_{\mathcal{X}} (x_t - \eta \nabla f(x_t)) \\ \text{Proximal GD : } x_{t+1} &= \text{prox}_{\eta h} (x_t - \eta \nabla f(x_t)) \end{aligned} \quad // \quad \begin{aligned} &\text{If } h(x) = \mathbb{I}_{\mathcal{X}}(x) \\ &\text{then } \text{prox}_{\eta h} = \text{Proj}_{\mathcal{X}} \end{aligned}$$

Each iteration requires:

- ① evaluation of ∇f
- ② evaluation of prox operator.

Examples of Prox Operators

$$f(x) = \|x\|_1 = \sum |x_i|$$

$$\text{prox}_{\eta f}(x) = \arg \min_u \|u\|_1 + \frac{1}{2\eta} \|u - x\|_2^2$$

$$u \text{ opt. sol'n to } \uparrow : 0 \in \partial \|u\|_1 + \frac{1}{\eta} (u - x)$$

$$\Leftrightarrow \boxed{(x - u) \in \eta \partial \|u\|_1}$$

$$\text{Recall: } z \in \partial \|u\|_1 \Rightarrow z_i = \begin{cases} \text{sign}(u_i) & \text{if } u_i \neq 0 \\ \in [-1, 1] & \text{if } u_i = 0 \end{cases}$$

$$(x - u) \in \eta \partial \|u\|_1 \Rightarrow u : u_i = \begin{cases} x_i - \eta & \text{if } x_i \geq \eta \\ 0 & \text{if } |x_i| \leq \eta \\ x_i + \eta & \text{if } x_i \leq -\eta \end{cases}$$

$$\text{Ex: } \text{prox}_f \begin{pmatrix} 2 \\ -1/2 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$$

"soft-thresholding"

"shrinkage"

Other examples of prox operator

$$\bullet \quad f(x) = \frac{1}{2} x^T Q x + q^T x + q_0 \quad (Q \succeq 0)$$

$$\text{prox}_{\eta f}(x) = (I + \eta Q)^{-1} (x - \eta q)$$

[check by using def'n's]

$$\bullet \quad f(x) = \sum f_i(x_i)$$

$$\Rightarrow (\text{prox}_{\eta f}(x))_i = \text{prox}_{\eta f_i}(x_i)$$

• Fact: Moreau Decomposition.

Fenchel - Transform: $f^*(y) = \sup_x \{ \langle y, x \rangle - f(x) \}$

$$x = \text{prox}_f(x) + \text{prox}_{f^*}(x)$$

Recall connection to projection.

Composite functions: smooth + easy-prox

$$\min: f(x) = \boxed{g(x)} + h(x)$$

↑ smooth ↑ easy-to-evaluate prox operator.

key motivation: Lasso

$$\min: f(x) = \underbrace{\frac{1}{2\eta} \|Ax - y\|_2^2}_{\text{smooth}} + \underbrace{\lambda_n \|x\|_1}_{\text{not smooth}}$$



Properties of Prox